

ALAN PETTIT

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Education

Technical University of Munich	2017-2019
M.Sc. Satellite Application Engineering	
University of Toronto	2012-2017
B.A.Sc. in Materials Science and Engineering (Honours)	
Minor in Engineering Business	

Technical Skills

Languages: English, German (C1), French (B2)

Experience: Mechanical engineering, precision opto-mechanics, space payload design

Programming: C, C++, Python, MATLAB

Software: Siemens Teamcenter/NX, Autodesk Vault/Inventor, SolidWorks, Hyperworks, Catia, Ansys, MS Office

Work Experience

Mynaric GmbH, Lead Space Mechanical Engineer	2022-present
- Coordinated the design, simulation, production and testing of an OCT telescope sub-assembly - Regularly presented work and status updates to internal and external stakeholders - Prioritized team tasks and tradeoffs to meet aggressive and evolving project deadlines	
Mynaric GmbH, Mechanical Development Engineer	2020-2022
- Designed opto-mechanical flight and GSE parts for satellite laser communication terminals - Produced technical drawings and design documentation	
IFEN GmbH, Masters Thesis	2019
- Simulated GNSS radio occultation events using raytracing - Implemented atmospheric and ionospheric models in C++ for software simulator	
Husky Injection Molding Systems Ltd., Metallurgy Intern	2015-2016
- Designed and performed experiments for quantifying the wear resistance of DLC coatings - Performed detailed failure analysis on machine components, using numerous lab techniques	

Project Experience

DEDRA Space Debris CubeSat, Communications Team Lead	2018-2019
- Iteratively designed a communications and ground segment to meet mission requirements - Attended the ESA Concurrent Engineering Workshop for CubeSats	
Human Powered Vehicles Design Team, Project Director	2014-2017
- Directed and managed a 30 member ASME HPVC engineering design team - Engineered, simulated, built and raced several composite streamlined bikes	

Awards

Hatch Engineering Aboriginal Scholarship	2014-2016
American Helicopter Society Igor I. Sikorsky Prize, Aerovelo team	2013